Craft 6: Inspect Your Own Work

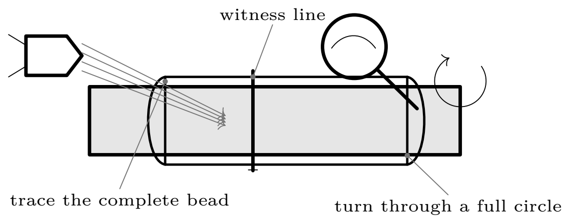
Shop skill — the one that makes the other five worth having

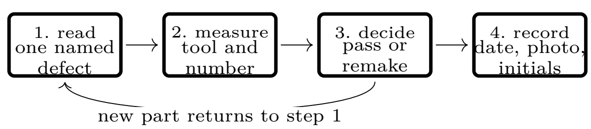
Confidence comes from evidence, not from hoping

THE STANDARD
Before anything is used, every measurement has been checked by someone, every joint has been looked at against a written list, every defect found has been **fixed or the part remade** — never noted and accepted — and the whole lot is recorded on paper with a date and a name against it.

Why this is a skill and not an attitude

Being careful is not a personality trait you either have or lack. It is a procedure: know what good looks like, look for specific named defects rather than a general impression, do it before the next step hides the evidence, and write down what you found. People who build reliable things are not more virtuous than people who do not — they have a list.





Look for named defects	"Does it look OK?" finds almost nothing, because your eye reports what you expect. "Is there a continuous bead all the way round this socket?" finds the actual problem. Every craft sheet ends with a check phrased that way for exactly this reason.
Inspect before you hide it	A burr is invisible once the pipe is in the socket; a weld bead is unreadable once the assembly is painted. Sequence the inspection before the step that conceals the thing.
Fix or remake	A defect you have written down and decided to live with is worse than one you never found, because now you are relying on a judgement you were not qualified to make. Pipe is cheap.
Someone else's eyes	Swap parts with your brother and check each other's. You cannot see your own mistakes well — your brain replays what you intended, not what you did.
Write it down	A build log turns "I think it cured long enough" into a date. *see the **Build Log** sheet*
Sign it	Putting your initials against a check is a small thing that changes how carefully the check gets done. That is not a trick; it is just true.

The inspection, in order

Run this before the assembly is accepted and again whenever it has been handled, stored or altered. Inspection is visual and dimensional only: do not use improvised pressure, flame or firing as a "proof" test.

	Check	Pass means
1	Cut faces	Square to the pipe, checked with a try square in four positions, no light. *see the **Craft 2: Cut Square** sheet*
2	Edges	No burr inside or out; an even 1–2 mm chamfer all the way round every pipe end. *see the **Craft 3: Deburr & Chamfer** sheet*
3	Every weld	A continuous squeeze-out bead with no gaps; witness marks aligned; pipe bottomed. *see the **Craft 4: Solvent Weld** sheet*
4	Cure	Forty-eight hours since the date written on the part. Not “about two days”.
5	Penetrations	Round holes, no crazing, no cracks, epoxy fillet continuous all the way round. *see the **Craft 5: Drill & Seal** sheet*
6	Chamber wall	No crazing, whitening, star cracks or deformation anywhere. Any of these retires the chamber permanently.
7	Cleanout	Threads clean and undamaged; mating parts seat by hand without force.
8	Igniter	Spark visible and audible across the gap, chamber open, before any fuel.
9	Barrel bore	Clean, smooth, nothing loose inside. Look down it *from the breech*, never from the muzzle.
10	Pipe marking	Every pressure-bearing part still legibly reads Schedule 40 pressure PVC with an NSF-PW rating.

Ask between every two checks

Question	Evidence required
What exactly did I inspect?	Part ID, location and the named defect from the row above.
What did the tool say?	A number with units, or a complete view under raking light; never “looks fine”.
What changed since last time?	Handling, storage, cure time, impact or alteration recorded explicitly.
Would another person reproduce my result?	Datum, tool and photograph are sufficient to repeat the check.
Is any evidence ambiguous?	If yes, quarantine and remake. Uncertainty is not a pass category.

ANY SINGLE FAILURE STOPS THE SESSION

There is no partial pass and no “we will watch it”. A chamber is a pressure vessel a few feet from your face; the whole value of an inspection list is that it decides in advance, calmly, what you will do when you find something — so that you are not making that call while three people stand around waiting to shoot.

Judging your own work honestly

Measure, do not eyeball	"That looks square" is not a check. The try square is a check.
Compare to a reference	Keep the sectioned practice weld from Craft 4 on the bench. Comparing against a known-good example is far more reliable than comparing against your memory of one.
Separate effort from quality	A joint you worked hard on is not thereby a good joint. The two are unrelated, and confusing them is the most common reason a defect gets argued away.
Expect to remake things	Everyone who builds well remakes parts. It is evidence the inspection is working, not evidence that you are bad at this. A build with no remade parts usually means nobody looked properly.
Do it when you are fresh	Inspection at the end of a long session finds less. If you are tired, write the log and inspect tomorrow.

The acceptance packet

Final proof is a completed ten-row checklist, the actual measurements with units, cure dates, photographs of every complete joint and penetration under raking light, and two sets of initials. Mark the assembly ACCEPTED only when every row passes. Mark any failed or uncertain part QUARANTINED, separate it physically, and remake it from new material. No firing or improvised pressure test can convert a failed inspection into a pass.

Check it yourself. Hand your finished assembly and this sheet to somebody else and ask them to run all ten checks without your help. Anything they find that you did not is worth more than anything you found yourself — it tells you which of your own checks you are performing loosely.

What you are really learning

Not how to make a potato launcher. How to know whether a thing you made is sound, and to be able to say why, and to be willing to cut a part up and do it again when it is not. That transfers to every workshop, every bike repair, every piece of code and every piece of furniture either of you ever makes. The launcher is a pretext.